Face Mask Detection

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***Abstract*— The importance of the monitoring of the usage of face masks has been realized during the COVID-19 pandemic. It is not possible to manually monitor the usage of face masks in public areas due to the presence of a large number of people. Manual monitoring can be erroneous. It is possible to apply the concepts of Artificial Intelligence and computer vision to solve the problem in an efficient way. The objective of the present project is to design an AI-based system for the purpose of detecting face masks. The present project is based on the application of deep learning concepts. The present project is based on the application of the concepts of Convolutional Neural Networks along with YOLOv8. The present project is based on the application of a dataset that is available in the public domain. The dataset contains images with face masks as well as images without face masks. The objective of the present project is to design an accurate and efficient system for the purpose of detecting face masks. The objective of the present project is to design an accurate and efficient AI-based system for the purpose of detecting face masks.**

Keywords*— *Face Mask Detection, Deep Learning, CNN, YOLOv8, Computer Vision, Image Classification

I. Introduction (*Heading 1*)

Due to the rapid spread of the COVID-19 pandemic in the world, the necessity for the development of an effective system in the fight against the spread of the disease in the world has been deemed to have significant importance. As recommended in the fight against the spread of the disease in the world, one of the proposals was the use of face masks. It has been difficult to monitor the use of face masks by individuals in crowded places such as airports, hospitals, and transport. However, the process has been more difficult when the process is done manually. However, with the recent developments in the application of AI and computer vision technologies, it has been possible to carry out different visual recognition processes automatically. Deep learning has been deemed to have the potential in the effective resolution of different computer vision-related classification problems. This has been exemplified by the use of CNN in the classification of different images with high accuracy. This has been exemplified by the use of CNN in the classification of images with or without face masks. YOLOv8 has been used in the detection of face images.

II. Literature review

A. Literature review1:

In the paper titled "Real-Time Facemask Detection for Preventing COVID-19 Spread Using Transfer Learning Based Deep Neural Network," the authors have proposed a deep learning model based on the concept of transfer learning. In this method, pre-trained neural networks have been used to increase the accuracy of the detection method..

B. Literature review2:

Another study titled "A Deep Learning-Based Approach for Real-Time Facemask Detection" presented a CNN-based system designed for real-time applications. The researchers developed a lightweight architecture capable of detecting mask usage from live video streams. Their model achieved high accuracy and demonstrated the potential of deep learning techniques for public safety monitoring.

C. Literature review3:

The paper "Face Mask Detection Using Convolutional Neural Networks" proposed a custom CNN model for classifying images of people wearing masks and not wearing masks. The study showed that CNN models can effectively extract facial features and identify mask usage with high reliability.

III. Dataset

For the purpose of this project, the dataset used is the one provided by Face Mask Detection Dataset. Dataset Link: https://www.kaggle.com/datasets/andrewmvd/face-mask-detection The dataset contains images of people wearing masks and images of people who are not wearing masks. The images in the dataset are labeled, i.e., it is specified whether the person is wearing the mask or not. Additionally, there are annotation files provided in the dataset, which can be used to find the face's position in an image. The operations to be performed on the dataset before training the model are as follows: Resizing the image Normalizing the pixels Splitting the dataset into the training dataset and the testing dataset.

Identify applicable funding agency here. If none, delete this text box.

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A. Methodology

The proposed system combines image classification and object detection techniques to detect face masks. First, the dataset is collected and preprocessed. Images are resized and normalized to prepare them for training. The dataset is then divided into training and testing sets. Next, a Convolutional Neural Network (CNN) is used to learn important visual features from the images. CNN layers automatically extract features such as edges, textures, and facial structures. These features are used to classify images into two categories: mask and no mask. In addition to CNN, the project incorporates YOLOv8, a modern object detection model that can identify and locate faces in real time. YOLOv8 detects the face region in an image or video frame and then the classification model determines whether the detected face is wearing a mask.

The overall workflow of the system is:

Dataset Collection → Image Preprocessing → CNN Training → YOLOv8 Detection → Mask Classification

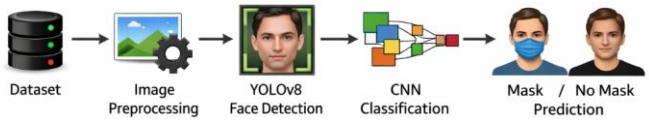
This combined approach allows the system to perform both accurate classification and real-time detection

Current Progress

At the midpoint of the semester, several important steps of the project have been completed. The literature review has been conducted to understand existing research on face mask detection using deep learning. Three relevant peer-reviewed research papers were studied to analyze different methodologies and approaches. The dataset for the project has been selected and downloaded. Initial preprocessing steps such as image resizing and dataset organization have been performed. The project architecture has also been designed, combining CNN for classification and YOLOv8 for real-time detection. Initial experimentation with the dataset and model setup is currently in progress. The next stage of the project will focus on training the model and evaluating its performance using standard evaluation metrics.

Expected Results

The expected outcome of this project is a deep learning model capable of accurately detecting whether a person is wearing a face mask. The model is expected to achieve high classification accuracy and demonstrate reliable detection performance in both image and video inputs. The system may also be extended to real-time camera applications for monitoring mask compliance in public spaces.



Future Work

In the final stage of the project, several improvements will be explored. The model will be trained and optimized to improve detection accuracy. Performance evaluation will be conducted using metrics such as accuracy, precision, recall, and F1-score. Future work may also include integrating the model with real-time camera systems and expanding the dataset to improve robustness in different lighting conditions and environments.

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